CHAPTER 2

The Plane

Definition

A plane is a surface such that if any two points are taken on it, the straight line joining them lies completely on the surface.

Art. 3.1

The general equation of first degree in x, y, and z, i.e.,

$$ax + by + cz + d = 0 \qquad (1)$$

represents a plane.

Corollary 1

General equation of a given plane that passes through a given point (x_1, y_1, z_1).

Since the plane (1) passes through the point (x_1, y_1, z_1), we get

$$ax_1 + by_1 + cz_1 = 0 \qquad (2)$$

Subtracting (2) from (1), we get

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0 \qquad (3)$$

Corollary 2

General equation of a plane that passes through the origin (0,0,0) is

$$ax + by + cz = 0 \qquad (4)$$

Art. 3.2

The equation of a plane passing through three given points (x_1, y_1, z_1), (x_2, y_2, z_2) and (x_3, y_3, z_3) is

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_2 & 1 \end{vmatrix} = 0$$

Art. 3.3 Standard forms of equations of planes

(a) **Intercept form:** If a plane passes through the points

$(a,0,0)$, $(0,b,0)$, and $(0,0,c)$, then

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

is the equation of a plane.

(b) **Normal form:** The equation of a plane in terms of p, where p is the length of the perpendicular from origin to the plane and l, m, n are the direction cosines of the perpendicular.

Let p be the length of the perpendicular ON from the origin O on the plane ABC, and l, m, n, the direction cosines of the normal ON to the plane ABC. Take any point $P(x,y,z)$ on the plane.

Then $\angle ONP = 90^0$, because ON is perpendicular to the plane and PN lies in the plane. Draw PL perpendicular to xz-plane and draw LM perpendicular to x-axis.

Then $ON =$ projection of OP on the line ON

= projection of OM + projection of LP + projection of ML

$$\Rightarrow p = lx + my + nz \qquad (1)$$

Thus, $lx + my + nz = p$ is the normal form to the equation of a plane.

Note

If coordinates of P are (x,y,z), then the projection of OP on a line with direction cosines l, m, n is $lx + my + nz$.

Corollary 1

In equation (1), we see that the coefficients of x, y, z are actual direction cosines of the normal to the plane.

Corollary 2

If $ax+by+cz+d=0$ represents the general equation of a plane, then the coefficients a, b, c will be the quantities proportional to the direction cosines of the normal to the plane.

Art. 3.4 Angle between two planes

Angle between two planes means the angle between the normal of the two planes from any point. Consider the planes

$$ax+by+cz=0 \qquad (1)$$

and

$$a_1x+b_1y+c_1z=0 \qquad (2)$$

If θ be the angle between the normal of the planes with direction ratios a,b,c and a_1,b_1,c_1 then

$$\cos\theta=\frac{aa_1+bb_1+cc_1}{\sqrt{a^2+b^2+c^2}\sqrt{a_1^2+b_1^2+c_1^2}} \qquad (3)$$

Art. 3.5

Condition of perpendicularity of the normal to the planes

If the planes (1) and (2) are mutually perpendicular, so the normal of the planes are also mutually perpendicular whose direction cosines are proportional to a,b,c and a_1,b_1,c_1. Hence the required condition is

$$aa_1+bb_1+cc_1=0 \tag{4}$$

Art. 3.6 Condition of parallelism of two planes

The planes (1) and (2) are parallel if and only if the coefficients of x, y, z in their equations are proportional, i.e., if

$$\frac{a}{a_1}=\frac{b}{b_1}=\frac{c}{c_1} \tag{5}$$

Art. 3.7 Some system of planes

(a) The equation of a plane which is parallel to the plane (1) and passing through the point (x_1,y_1,z_1) is

$$a(x-x_1)+b(y-y_1)+c(z-z_1)=0 \tag{6}$$

(b) The equation of a plane which is perpendicular to the plane (1) and passing through the point (x_1,y_1,z_1) and (x_2,y_2,z_2) is

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ a & b & c & 1 \end{vmatrix}=0 \text{ or } \begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_1-x_2 & y_1-y_2 & z_1-z_2 \\ a & b & c \end{vmatrix}=0 \tag{7}$$

(c) Planes perpendicular to coordinate planes

(i) The equation of the plane perpendicular to *yz*-plane is
$$by+cz+d=0 \tag{8}$$
(ii) The equation of the plane perpendicular to *xz*-plane is
$$ax+cz+d=0 \tag{9}$$
(iii) The equation of the plane perpendicular to *xy*-plane is
$$ax+by+d=0 \tag{10}$$

Note

The plane $ax+by+cz+d=0$ represents planes perpendicular to *yz*-, *zx*-, or *xy*- planes according as $a=0$, $b=0$ or $c=0$.

(d) Plane through the line of intersection of the planes (1) and (2) is

$$ax+by+cz+d+\lambda(a_1x+b_1y+c_1z+d)=0 \qquad (11)$$

where λ is an arbitrary constant which may be obtained if the plane (11) is parallel to the given plane.

Solved Problems

1. Find the equation of the plane passing through the points (2,3,1), (1,1,3) and (2,2,3). Find also the perpendicular distance from the point (5,6,7) to the plane.

Solution: Any plane passing through the point (2,3,1) is

$$a(x-2)+b(y-3)+c(z-1)=0 \qquad (1)$$

Since (1) passes through the points (1,1,3) and (2,2,3), we get

$$a(1-2)+b(1-3)+c(3-1)=0$$

$$\Rightarrow a+2b-2c=0 \qquad (2)$$

and $a(2-2)+b(2-3)+c(3-1)=0$

$$\Rightarrow b-2c=0 \qquad (3)$$

From (2) and (3), we get

$$\frac{a}{-4+2}=\frac{b}{0+2}=\frac{c}{1-0} \Rightarrow \frac{a}{-2}=\frac{b}{2}=\frac{c}{1}=k, \text{ say} \qquad (4)$$

Hence the required equation of the plane is given by

$$-2k(x-2)+2k(y-3)+k(z-1)=0$$

$$\Rightarrow\ 2x-2y-z+3=0 \qquad (5)$$

Second Part

The perpendicular distance from (5,6,7) to the plane (5) is

$$\Rightarrow\ \frac{|(2)(5)-(2)(6)-7+3|}{\sqrt{4+4+1}}=\frac{6}{3}=2.$$

Corollary 1

The distance of (x_1,y_1,z_1) from the plane $ax+by+cz+d=0$ if the axes are rectangular, is given by $\dfrac{ax_1+by_1+cz_1+d}{\pm\sqrt{a^2+b^2+c^2}}$. If d is positive then the positive sign is to be taken, as it gives a positive value for the perpendicular from the origin.

Corollary 2

If d is positive, the expression $ax_1+by_1+cz_1+d$ is positive if (x_1,y_1,z_1) and the origin are on the same side of the plane $ax+by+cz+d=0$, and negative if they are on opposite sides.

2. Show that the four points (0,−1,−1), (4,5,1), (3,9,4) and (−4,4,4) are coplanar.

Solution: Equation of any plane through the points (0,−1,−1), (4,5,1), (3,9,4) is

$$5x-7y+11z+4=0 \qquad (6)$$

Since the point (−4,4,4) satisfies equation (6), the given points are coplanar, i.e., they lie in the same plane.

3. Find the equation of the plane passing through the intersection of the planes

$$x+2y+3z+4=0 \text{ and } 4x+3y+2z+1=0$$

and the point (1,2,3).

Solution: Given planes are $x+2y+3z+4=0$ (7)

and $4x+3y+2z+1=0$ (8)

Any plane passing through the line of intersection of the planes (7) and (8) is

$$x+2y+3z+4+\lambda(4x+3y+2z+1)=0 \qquad (9)$$

where λ is an arbitrary constant.

Since (9) passes through the point (1,2,3), we get

$$1+2\times2+3\times3+4+\lambda(4\times1+3\times2+2\times3+1)=0 \Rightarrow \lambda=-\frac{18}{17}$$

Hence the equation of the required plane is

$$x+2y+3z+4+\left(-\frac{18}{17}\right)(4x+3y+2z+1)=0$$

$$\Rightarrow 17x+34y+51z+68-72x-54y-36z-18=0$$

$$\Rightarrow 11x+4y-3z-10=0$$

4. Find the angle between the planes $2x-y+z=6$ and $x+y+2z=7$.

Solution: Let θ be the angle between the planes. Then

$$\cos\theta=\frac{(2)(1)+(-1)(1)+(1)(2)}{\sqrt{2^2+(-1)^2+1^2}\sqrt{1^2+1^2+2^2}}=\frac{3}{6}=\frac{1}{2}=\cos\frac{\pi}{3}$$

Hence $\theta=\frac{\pi}{3}$.

5. Find the equation of the plane passing through the point $(4,-2,1)$ and parallel to the plane whose direction ratios are $7,2,-3$.

Solution: Any plane through the point $(4,-2,1)$ is

$$a(x-4)+b(y+2)+c(z-1)=0 \qquad (1)$$

Since (1) is parallel to the plane whose direction ratios are $7,2,-3$, we get

$$\Rightarrow \frac{a}{7}=\frac{b}{2}=\frac{c}{-3}=\lambda, \text{ (say)} \Rightarrow a=7\lambda,\ b=2\lambda,\ c=-3\lambda$$

Hence the required equation of the plane is

$$7\lambda(x-4)+2\lambda(y+2)-3\lambda(z-1)=0$$

$$\Rightarrow 7x+2y-3z-21=0$$

CHAPTER 2

The Plane

Solved Problems

6. Find the equation of the plane passing through the point (4,0,1) and parallel to the plane $4x+3y-12z+6=0$.

Solution: The equation of the parallel planes differ by a constant only. Hence the equation of the plane parallel to $4x+3y-12z+6=0$ is

$$4x+3y-12z+\lambda=0 \qquad (1)$$

Since (1) passes through the point (4,0,1), we get

$(4)(4)+(3)(0)-(12)(1)+\lambda=0 \Rightarrow \lambda=-4$

Hence the required equation of the plane is

$$4x+3y-12z-4=0.$$

7. Find the equation to the plane passing through the line of intersection of the planes $2x-y=0$ and $3z-y=0$ and perpendicular to the plane $4x+5y-3z+1=0$.

Solution: Given planes are $2x-y=0 \qquad (1)$

$$3z-y=0 \qquad (2)$$

$$4x+5y-3z+1=0 \qquad (3)$$

Any plane passing through the line of intersection of the planes (1) and (2) is

$$2x-y+\lambda(3z-y)=0 \Rightarrow 2x-(1+\lambda)y+3\lambda z=0 \qquad (4)$$

where λ is an arbitrary constant.

Since (3) and (4) are perpendicular, we get

$$(4)(2)+5(-1-\lambda)-3(3\lambda)=0 \Rightarrow \lambda=\frac{3}{14}$$

Hence the required equation of the plane is

$$2x-y+(3/14)(3z-y)=0 \Rightarrow 28x-17y+9z=0.$$

8. Find the equation to the plane passing through the points (2,2,1) and (9,3,6) and perpendicular to the plane $2x+6y+6z=9.$

Solution: Any plane through the point (2,2,1) is

$$a(x-2)+b(y-2)+c(z-1)=0 \qquad (1)$$

Since (1) passes through the point (9,3,6), we get

$$a(9-2)+b(3-2)+c(6-1)=0 \Rightarrow 7a+b+5c=0 \qquad (2)$$

Since (1) is perpendicular to the plane $2x+6y+6z=9$, we get

$$2a+6b+6c=0 \qquad (3)$$

From (2) and (3), we get

$$\frac{a}{6-30}=\frac{b}{10-42}=\frac{c}{42-2}$$

$$\Rightarrow \frac{a}{-24}=\frac{b}{-32}=\frac{c}{40} \Rightarrow \frac{a}{3}=\frac{b}{4}=\frac{c}{-5}=k, \text{ say.}$$

Hence the required equation of the plane is

$$3k(x-2)+4k(y-2)-5k(z-1)=0$$

$$\Rightarrow 3x+4y-5z=9$$

9. Show that the equation to the plane passing through $P(2,3,-1)$ at right angles to *OP* is $2x+3y-z=14$.

Solution: The direction cosines of *OP* are

$$\frac{2-0}{\sqrt{4+9+1}},\frac{3-0}{\sqrt{4+9+1}},\frac{-1-0}{\sqrt{4+9+1}} \text{ or } \frac{2}{\sqrt{14}},\frac{3}{\sqrt{14}},\frac{-1}{\sqrt{14}}$$

Any plane at right angles to *OP* is

$$\frac{2}{\sqrt{14}}x+\frac{3}{\sqrt{14}}y-\frac{1}{\sqrt{14}}z=p, \text{ where } p=\sqrt{14}.$$

Hence the required equation of the plane is

$$2x+3y-z=14.$$

10. A plane meets the coordinate axes in *A, B, C* such that the centroid of the triangle *ABC* is the point *p, q, r*; show that the equation of the plane is

$$\frac{x}{p}+\frac{y}{q}+\frac{z}{r}=3.$$

Solution: Let the equation of the plane be

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1 \qquad (1)$$

So the coordinates of *A, B, C* are $(a,0,0)$, $(0,b,0)$, $(0,0,c)$, respectively, and the centroid of the triangle *ABC* is $G\left(\frac{a}{3},\frac{b}{3},\frac{c}{3}\right)$. According to the question, $p=\frac{a}{3},\ q=\frac{b}{3},\ r=\frac{c}{3}$.

Hence the required equation of the plane is

$$\frac{x}{3p}+\frac{y}{3q}+\frac{z}{3r}=1 \quad \Rightarrow \frac{x}{p}+\frac{y}{q}+\frac{z}{r}=3.$$

11. A variable plane is at a constant distance p from the o origin and meets the axes in A, B, C. Show that the locus of the centroid of the tetrahedron $OABC$ is

$$x^{-2}+y^{-2}+z^{-2}=16p^{-2}.$$

Solution: Let the equation of the plane be

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1 \qquad (1)$$

So the coordinates of A, B, C are $(a,0,0)$, $(0,b,0)$, $(0,0,c)$, respectively.

Therefore, the centroid of the tetrahedron $OABC$ is $G\left(\frac{a}{4},\frac{b}{4},\frac{c}{4}\right)$.

So, $x=\frac{a}{4},\ y=\frac{b}{4},\ z=\frac{c}{4} \Rightarrow a=4x,\ b=4y,\ c=4z$.

The distance of (1) from the origin is

$$p=\frac{1}{\sqrt{(1/a)^2+(1/b)^2+(1/c)^2}}$$

$$\Rightarrow \frac{1}{p^2}=\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}$$

Hence the required equation of the plane is

$$\frac{1}{p^2}=\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2} \Rightarrow \frac{1}{p^2}=\frac{1}{(4x)^2}+\frac{1}{(4y)^2}+\frac{1}{(4z)^2}$$

$$\Rightarrow x^{-2}+y^{-2}+z^{-2}=16p^{-2}.$$

12. A variable plane is at a constant distance *p* from the origin and meets the axes in *A, B, C*. Through *A, B, C* planes are drawn parallel to the coordinate planes. Show that the locus of their point of intersection is

$$x^{-2}+y^{-2}+z^{-2}=p^{-2}.$$

Solution: Let the equation of the plane be

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1 \qquad (1)$$

So the coordinates of *A*, *B*, *C* are (a,0,0), (0,b,0), (0,0,c), respectively.

Therefore, the equation of the planes through *A*, *B*, *C* and parallel to the coordinate planes are $x=a$, $y=b$ and $z=c$.

The distance of (1) from the origin is

$$p=\frac{1}{\sqrt{(1/a)^2+(1/b)^2+(1/c)^2}}$$

$$\Rightarrow \frac{1}{p^2}=\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}$$

Hence the required equation of the plane is

$$\frac{1}{p^2}=\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2} \quad \Rightarrow \frac{1}{p^2}=\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}$$

$$\Rightarrow x^{-2}+y^{-2}+z^{-2}=p^{-2}.$$

13. A variable plane passes through a fixed point (a,b,c) and meets the coordinate axes in *A*, *B*, *C*. Show that the locus of the point common to the plane through *A*, *B*, *C* parallel to the coordinate planes is

$ayz + bxz + cxy = xyz.$

Solution: Let the equation of the plane be

$$\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1 \qquad (1)$$

So the coordinates of *A*, *B*, *C* are $(\alpha,0,0)$, $(0,\beta,0)$ and $(0,0,\gamma)$, respectively.

Since it passes through (a,b,c), we have

$$\frac{a}{\alpha} + \frac{b}{\beta} + \frac{c}{\gamma} = 1 \qquad (2)$$

Now, the equation of the planes through *A*, *B*, *C* and parallel to the coordinate planes are $x = \alpha$, $y = \beta$ and $z = \gamma$.

Hence the locus of their point of intersection will be obtained by eliminating α, β, γ from equation (2):

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 1 \quad \text{or} \quad ayz + bxz + cxy = xyz.$$

14. A point P moves on the plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ which is fixed. The plane through P perpendicular to OP meets the axes in A, B, C. The plane through A, B, C parallel to yz, zx, xy planes intersect in Q. Prove that if the axes be rectangular, the locus of Q is

$$\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{1}{ax}+\frac{1}{by}+\frac{1}{cz}$$

Solution: Given the fixed plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ (1)

Let the coordinates of the variable point P be (α,β,γ).

Since the plane (1) contains P, we get

$$\frac{\alpha}{a}+\frac{\beta}{b}+\frac{\gamma}{c}=1 \qquad (2)$$

Now the direction ratios of OP are

$\alpha-0, \beta-0, \gamma-0$ or α, β, γ

Also, since the direction cosines of OP are proportional to $\alpha, \beta, \gamma,$ so these are also direction ratios of the normal to the plane which is perpendicular to OP.

Hence any plane through P and perpendicular to OP is

$\alpha(x-\alpha)+\beta(y-\beta)+\gamma(z-\gamma)=0$

$\Rightarrow \alpha x+\beta y+\gamma z=\alpha^2+\beta^2+\gamma^2$ (3)

It meets the axes in A, B, C whose coordinates are respectively

$$\left((\alpha^2+\beta^2+\gamma^2)/\alpha,0,0\right),\ \left(0,(\alpha^2+\beta^2+\gamma^2)/\beta,0\right),$$

$$\left(0,0,(\alpha^2+\beta^2+\gamma^2)/\gamma\right)$$

So the equation of the plane through *A* and parallel to the yz-plane is $x=(\alpha^2+\beta^2+\gamma^2)/\alpha$.

Similarly, the equation of the plane through *B* and parallel to the xz-plane is $y=(\alpha^2+\beta^2+\gamma^2)/\beta$ and the equation of the plane through *C* and parallel to the *xy*-plane is $z=(\alpha^2+\beta^2+\gamma^2)/\gamma$.

Now,

$$\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}$$

$$=\left(\frac{\alpha}{\alpha^2+\beta^2+\gamma^2}\right)^2+\left(\frac{\alpha}{\alpha^2+\beta^2+\gamma^2}\right)^2+\left(\frac{\alpha}{\alpha^2+\beta^2+\gamma^2}\right)^2$$

$$=\frac{\alpha^2+\beta^2+\gamma^2}{(\alpha^2+\beta^2+\gamma^2)^2}=\frac{1}{\alpha^2+\beta^2+\gamma^2}$$

and

$$\frac{1}{ax}+\frac{1}{by}+\frac{1}{cz}$$

$$=\frac{\alpha/a}{\alpha^2+\beta^2+\gamma^2}+\frac{\beta/b}{\alpha^2+\beta^2+\gamma^2}+\frac{\gamma/c}{\alpha^2+\beta^2+\gamma^2}$$

$$=\frac{1}{\alpha^2+\beta^2+\gamma^2}$$

$$=\left(\frac{\alpha/a+\beta/b+}{\alpha^2+\beta^2+\gamma^2}\right)^2+\left(\frac{\beta/b}{\alpha^2+\beta^2+\gamma^2}\right)^2+\left(\frac{\gamma/c}{\alpha^2+\beta^2+\gamma^2}\right)^2$$

$$=\frac{1}{\alpha^2+\beta^2+\gamma^2} \text{ since } \frac{\alpha}{a}+\frac{\beta}{b}+\frac{\gamma}{c}=1$$

This completes the proof.

15. A variable plane makes intercepts on the coordinate axes, the sum of whose squares is constant. Find the locus of the foot of the perpendicular from the origin to the plane.

Solution: Let the equation of the plane be

$$ax+by+cz=d \quad (1)$$

or $$\frac{x}{d/a}+\frac{y}{d/b}+\frac{z}{d/c}=1 \quad (2)$$

So, (2) meets the coordinate axes in *A, B, C* whose coordinates are respectively $(d/a,0,0)$, $(0,d/b,0)$ and $(0,0,d/c)$.

Given that

$(d/a)^2+(d/b)^2+(d/c)^2=k^2$, say (3)

Let (α,β,γ) be the foot of the perpendicular from the origin to the plane. So, the direction cosines of the perpendicular are proportional to (α,β,γ).

Since (1) passes through (α,β,γ), we get

$$a\alpha+b\beta+c\gamma=d \quad (4)$$

Again, since the direction cosines of the normal to the plane (1) are proportional to a,b,c, we can write

$$\frac{\alpha}{a}=\frac{\beta}{b}=\frac{\gamma}{c} \Rightarrow \frac{\alpha^2}{a\alpha}=\frac{\beta^2}{b\beta}=\frac{\gamma^2}{c\gamma}=\frac{\alpha^2+\beta^2+\gamma^2}{a\alpha+b\beta+c\gamma}$$

$$=\frac{\alpha^2+\beta^2+\gamma^2}{d} \qquad (5)$$

So that $\dfrac{d}{a}=\dfrac{\alpha^2+\beta^2+\gamma^2}{\alpha}$, $\dfrac{d}{b}=\dfrac{\alpha^2+\beta^2+\gamma^2}{\beta}$,

$$\frac{d}{c}=\frac{\alpha^2+\beta^2+\gamma^2}{\gamma}$$

Thus, from (3), we get

$$\left(\frac{\alpha^2+\beta^2+\gamma^2}{\alpha}\right)^2+\left(\frac{\alpha^2+\beta^2+\gamma^2}{\beta}\right)^2+\left(\frac{\alpha^2+\beta^2+\gamma^2}{\gamma}\right)^2=k^2$$

$$(\alpha^2+\beta^2+\gamma^2)2\left(\frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2}\right)=k^2$$

Putting $x=\alpha, y=\beta$ and $z=\gamma$, we get

$$(x^2+y^2+z^2)2\left(\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}\right)=k^2$$

which is the required locus.

Exercises

1. Show that the four points (1,1,0), (-2,2,1), (1,2,-1) and (1,0,-1) are coplanar.
2. Prove that the line joining the points (2,2,-1) and (3,4,2) intersects the line joining the points (7,0,6) and (2,5,1).
3. Find the equation to the plane passing through the intersection of the planes $x+2y+3z+4=0$ and $4x+3y+2z+1=0$ and perpendicular to the plane $x+y+z+9=0$. Also show that it is perpendicular to the *xz*-plane.
4. Find the distance of the point (2,0,1) and (3,-3,2) from the plane $x-2y+z=6$ and find whether the points lie on the same side or opposite side of the plane.
5. Find the equation to the plane perpendicular to the *yz*-plane and passing through the points (1,-2,4) and (3,-4,5).